The Sphere Packing Problem in dimensions 8 and 24: an overview of the solution

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Abstract

The goal of this article is to give an overview of Viazovska's solution of the sphere packing problem in dimensions 8 and 24. We introduce and prove some general results about the sphere packing problem and modular forms. Cohn and Elkies linear programming bounds are discussed and we work backwards in Viazovska's proof to give the main ideas to the reader.

Contents

1 Introduction **2**

2 Sphere packing generalities **2**

2.1 Lattice packings and periodic packings 3

2.2 The E_8 lattice and the Leech lattice 4

3 Linear Programming Bounds **5**

4 Modular Forms **8**

4.1 Motivation to the study of modular forms 8

4.2 The full modular group 9

4.3 Modular forms: generalities 11

4.4 Modular forms over $\Gamma(1)$ 14

4.5 Quasimodular forms 15

4.6 Theta series 16

5 An overview of Viazovska's proof **16**

5.1 The $+1$-eigenfunction 17

5.2 The -1-eigenfunction 22

6 Formal proof **23**

6.1 Verifying previous steps 23

6.2 Computational assistance 27

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1 Introduction

The sphere packing problem asks what is the densest way one can pack spheres of equal radius that do not overlap in $\mathbb{R}^d$. Although it may seem a naive question, it turns out to be one of hardest problems in geometry. The problem's first appearance was in 1611, when Kepler conjectured that the densest way of packing cannonballs was using the face-centred cubic packing. Yet, it was solved in a non-trivial dimension only 300 years later, in 1940: Fejes Tóth proved that the hexagonal lattice gives an optimal packing in $\mathbb{R}^2$. In 1998, Thomas Hales solved Kepler's conjecture, using an approach suggested by Tóth. The proof is extremely complex, use extensively computer assistance and was only fully computer verified in 2014. Until 2016, the answer for the sphere packing problem was only known in dimensions 1, 2 and 3. More historical facts of the problem can be found in [13].

In 2003, Cohn and Elkies, in [4], established a beautiful connection between the sphere packing problem and a Fourier uncertainty one, the so-called linear programming bounds. Their result asserts that the construction of a *magic function* satisfying some properties gives a bound on the sphere packing constant. They observed on numerics that the bounds in dimensions 2, 8 and 24 were really close to the density of known lattices packings and conjectured that this approach could solve the problem in these dimensions.

In 2016, Maryna Viazovska figured out how to construct the magic function that solves the sphere packing problem in dimension 8, by cleverly using the machinery of modular forms (see [14]). Only one week later, Viazovska jointly with Henry Cohn, Abhinav Kumar, Stephen Miller and Danylo Radchenko in [3] also solved the problem in dimension 24, using similar techniques. In this article, we work backwards in Viazovska's proof and deduce how the magic functions look like, starting from Cohn and Elkies linear programming bounds. The article is organized as follows:

In Section 2, a formal statement of the sphere packing problem is given, some general results are presented and the packing that solves the problem in dimension 8 is constructed. We do not give a definition of the best packing in dimension 24, in view of the its intricate formulation. However, we state some of its nice properties. We follow [6] to write this section.

In Section 3, Cohn and Elkies' linear programming bounds is proved. We also start our search for the magic function by deducing some of the properties it would have to satisfy in case it existed.

We motivate the study of modular forms in Section 4. The main features of the full modular group and modular forms are stated. Afterwards, we study Eisenstein series and theta series. The main reference for this section is [1].

At last, in Section 5, we focus ourselves in dimension 8, for simplicity, and give an overview of Viazovska's proof. We find a candidate for the magic function and, in Section 6, we show that it does solve the problem in dimension 8. The results are based on [2] and [11].

2 Sphere packing generalities

We define a sphere packing $\mathcal{P}$ as a collection of disjoint congruent balls in $\mathbb{R}^d$ that do not overlap. To be able to compare packings and characterize which is the best packing, we introduce

the notion of density. The density of a packing $\mathcal{P}$ is given by

$$\Delta_{\mathcal{P}} = \limsup_{r\to\infty} \frac{vol(B_r^d(0) \cap \mathcal{P})}{vol(B_r^d(0))},$$

where $B_r^d(x)$ denotes the ball in $\mathbb{R}^d$ centred in x and with radius r. We omit the x in $B_r^d(x)$ when it is irrelevant.

Finding the densest packing in each dimension is extremely hard, but one could try to give bounds to the constant

$$\Delta_d = \sup_{\mathcal{P}} \Delta_{\mathcal{P}},$$

where the supremum is taken over all sphere packings in $\mathbb{R}^d$. We call it the sphere packing constant in dimension d. A question that arises naturally is whether the sphere packing constant is attained by some packing $\mathcal{P}$ in every dimension. The answer is positive and was given by H. Groemer [8]. However, packings that have maximal density can be quite irregular and, consequently, are very hard to work with. Thus, we focus our attention on packings that have some regularity.

2.1 Lattice packings and periodic packings

A lattice Λ in $\mathbb{R}^d$ is a discrete subgroup of full rank. That is, Λ is given by linear combinations with integer coefficients of a basis of $\mathbb{R}^d$. Naturally, we define a lattice packing as a sphere packing whose spheres are centred in lattice points.

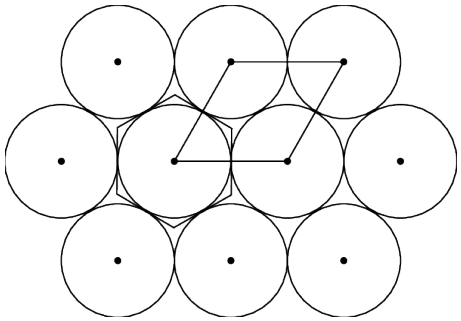


Figure 1: Lattice packing in $\mathbb{R}^2$: the hexagonal sphere packing.

It is easier to compute the density of these packings: if $\mathcal{P}$ is a lattice packing over the lattice Λ with balls of radius r, then

$$\Delta_{\mathcal{P}} = \frac{vol(B_r^d(0))}{vol(\mathbb{R}^d/\Lambda)},$$

where $vol(\mathbb{R}^d/\Lambda)$ is the volume of fundamental parallelogram of the lattice.

In the five dimensions which the sphere packing problem was solved, the extremizers are given by lattice packings that have special properties. However, it is not expected that lattice packings can solve the problem generally. Even more, we do not hope that lattice packings can have density arbitrarily close to Δ_d. For instance, in dimension 10 the density of the best

packing known is 8% greater than density of the best lattice packing known, as observed in [2]. Therefore, we allow more irregularity to get better properties.

We define a periodic packing as a sphere packing whose centres of the spheres are given by a finite translates of a lattice. More specifically, given a periodic packing $\mathcal{P}$, there exists a lattice Λ and points x_1, ..., x_m in $\mathbb{R}^d$, such that $\mathcal{P}$ can be written as

$$\mathcal{P} = \bigcup_{i=1}^{m} \bigcup_{v \in \Lambda} x_i + v + B_r^d(0).$$

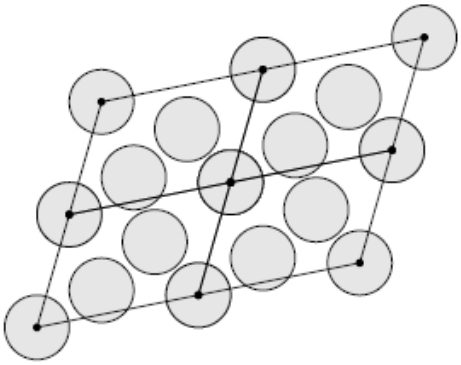


Figure 2: A periodic packing that is not a lattice packing.

It is also easier to compute the density of a periodic packing. If $\mathcal{P}$ is as above, it holds that

$$\Delta_{\mathcal{P}} = \frac{m \cdot Vol(B_r^d(0))}{Vol(\mathbb{R}^d/\Lambda)}.$$

Although it is not believed that lattice packings can approximate the densest sphere packing in each dimension, we know that this holds for periodic packings. More precisely, given any packing $\mathcal{P}$, $\Delta_{\mathcal{P}}$ can be approximated arbitrarily well by a sequence $\Delta_{\mathcal{P}_n}$, where $\mathcal{P}_n$ are periodic packings. The proof of this result can be found in [11, pg. 9].

2.2 The E_8 lattice and the Leech lattice

The E_8 lattice and the Leech lattice Λ_{24} are the lattices that generate the optimal sphere packings in dimensions 8 and 24, respectively. They are very special lattices because of their symmetry properties: they are even and unimodular. This special kind of lattice appears only in high dimensions and one can prove that the first dimension that they appear is 8. The E_8 lattice was first explicitly constructed in 1873, by Korkine and Zolotoreff. Here we give a simple construction of the E_8 lattice and prove some of its properties. The construction of the Leech lattice is more convoluted, so we restrict ourselves to state some of its properties.

The E_8 lattice can be defined as $E_8 = \{(x_1, \ldots, x_8); x_i \in \mathbb{Z} \text{ or } x_i \in \mathbb{Z} + \frac{1}{2}; \sum x_i \equiv 0 \bmod 2\}$. It is not hard to see that this definition coincides with the one given by $\mathbb{Z}^8 A$, where A is the matrix

$$A = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

We denote by v_i the i-th row vector of A. One can notice that this lattice is unimodular, since $\det(A) = 1$. We also observe that $|v_i|^2$ is even, for $i = 1, ..., 8$, thus the lattice is an even lattice. It is simple to observe that this implies that E_8 is a self-dual lattice.

We now compute its minimal norm, that is $\inf\{|v|; v \in E_8\}$. First, we notice that $|v_2| = |v_3| = \cdots = |v_8| = \sqrt{2}$ and $|v_1| = 2$. We separate in four cases to show that $|\alpha v_i + \beta v_j|^2 \geq 2$, for all $\alpha, \beta \in \mathbb{Z}$ and $i, j = 1, \ldots, 8$. Observe that if $\alpha = 0$ or $\beta = 0$ the result is trivial. Therefore, we consider $\alpha, \beta \in \mathbb{Z} \setminus \{0\}$ in the following cases:

1. For $i, j = 2, \ldots, 7$, $i < j$, it holds that $|\alpha v_i + \beta v_j|^2 \geq \alpha^2 + \beta^2 \geq 2$, since we have a $-\alpha$ and a β in different coordinates;

2. For $i = 2, \ldots, 7$ we analyse the norm of $\alpha v_i + \beta v_8$. We know that this vector has a coordinate equal to $-\alpha + \frac{\beta}{2}$ and other equal to $\alpha + \frac{\beta}{2}$. Thus, $|\alpha v_i + \beta v_8|^2 \geq (\alpha + \frac{\beta}{2})^2 + (-\alpha + \frac{\beta}{2})^2 > 2\alpha^2 \geq 2$;

3. It is easy to see that for $i = 3, \ldots, 7$, $|\alpha v_1 + \beta v_i|^2 > 2$. For $i = 2$ we have $|\alpha v_1 + \beta v_3|^2 = (2\alpha - \beta)^2 + \beta^2$. If $2\alpha = \beta$, then $|\alpha v_1 + \beta v_3|^2 = (2\alpha)^2 > 2$. Otherwise, the inequality follows trivially;

4. Finally, observe that $|\alpha v_1 + \beta v_8|^2 = (2\alpha + \frac{\beta}{2})^2 + 7(\frac{\beta}{2}) \geq 8\frac{\beta^2}{4} \geq 2$.

This proves that the minimal norm of E_8 is $\sqrt{2}$. Consequently, the sphere packing of balls of radius $\sqrt{2}/2$ centred in E_8 have density $vol\left(B^8_{\frac{\sqrt{2}}{2}}\right)$.

The construction of the Leech lattice can be found in [6, Chap. 4]. This lattice is also unimodular an even and, consequently, self-dual. It is the unique lattice in dimension 24 that has maximal minimal norm, which is equal to 2. Therefore, the sphere packing of balls of radius 1 centred in Λ_{24} has density equals to $vol\left(B^{24}_1\right)$.

3 Linear Programming Bounds

Although the proofs of the sphere packing problem in dimensions 2 and 3 are convoluted, Viazovska's proofs in dimensions 8 and 24 are quite simple and the reason for that is Cohn and Elkies' linear programming bounds. Cohn and Elkies' result asserts that the construction a *magic function* f satisfying some properties give bounds on the sphere packing constant. Precisely,

Theorem 3.1. *Let $f : \mathbb{R}^d \to \mathbb{R}$ be a Schwartz function. Assume that f satisfies*

$$f(0) = \hat{f}(0) = 1$$
$$\hat{f}(x) \geq 0, \forall x \in \mathbb{R}^d,$$

and that there exists a $r > 0$ such that $f(x) \leq 0$ for all $|x| \geq r$. Then,

$$\Delta_d \leq vol(B^d_{r/2}(0)).$$

Proof. We let $\mathcal{P}$ be a periodic packing of balls of radius $r/2$ (note that we can assume that, since one could dilate the packing if necessary). Hence, $\mathcal{P}$ can be written as

$$\mathcal{P} = \bigcup_{i=1}^{m} \bigcup_{v \in L} v + x_i + B^d_{r/2}(0),$$

where $x_i \in \mathbb{R}^d$, $i = 1, \ldots, m$, and L is a lattice.

Applying Poisson Summation formula for f we get that

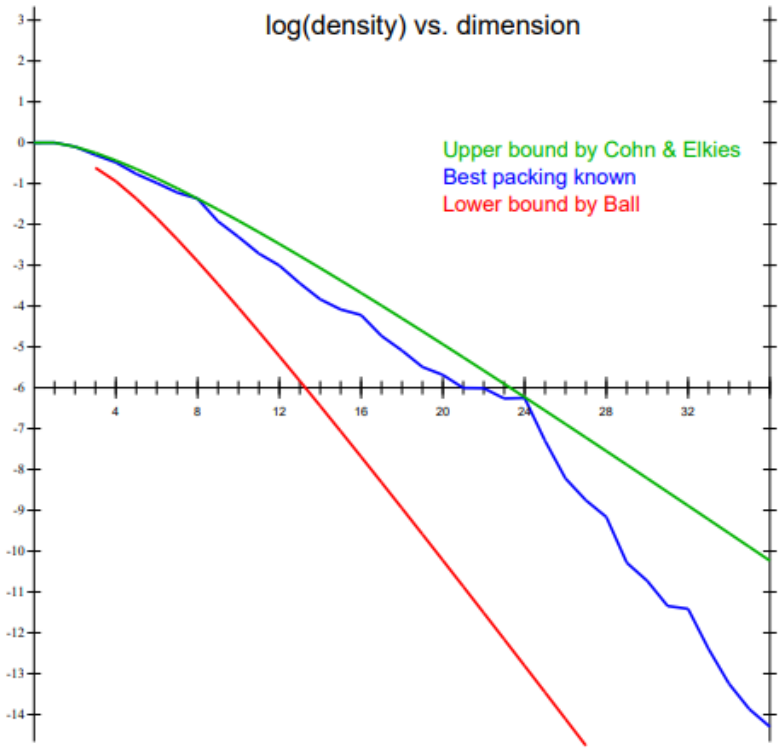
$$\sum_{v \in L} \sum_{i,j=1}^{m} f(v + x_j - x_i) = \frac{1}{vol(\mathbb{R}^d/L)} \sum_{u \in L^*} \sum_{i,j=1}^{m} \hat{f}(u) e^{2\pi i u \cdot (x_j - x_i)} = \frac{1}{vol(\mathbb{R}^d/L)} \sum_{u \in L^*} \hat{f}(u) \left| \sum_{i=1}^{m} e^{2\pi i u \cdot x_i} \right|^2. \tag{3.1}$$

We observe that the left hand side of 3.1 is smaller or equal to $mf(0)$, since $|v + x_i - x_j| \geq r$, for $i, j \neq 0$. On the other hand, the right hand side of the equation is greater or equal to $m^2 \hat{f}(0)$, since $\hat{f}$ is always non-negative. Therefore, we conclude that $1 \geq \frac{m}{vol(\mathbb{R}^d/L)}$ and by multiplying the equation by $vol(B_d(r/2))$ the desired result follows.

□

In the statement of the theorem we assumed that f is a Schwartz function, since we only deal with these functions in our work. However, the proof holds with more general hypothesis, for functions f such that Poisson Summation Formula can be applied.

Cohn and Elkies ran numerics using functions given by polynomials times Gaussians, and realized that the bounds in dimensions 8 and 24 were really close to the density of the best packing known. They conjecture that this method could solve the sphere packing problem in these dimensions, but until 2016 no one knew how to construct the magic functions.



By assuming that such magic functions exist, one can easily deduce some properties that they have to satisfy. For instance, suppose that there exists a Schwartz function $f : \mathbb{R}^8 \to \mathbb{R}$ such that the hypothesis of theorem 3.1 are verified for $r = \sqrt{2}$. This would imply that $\Delta_8 = \Delta_{E_8}$, proving that the packing generated by the E_8 lattice has the greatest density possible in dimension 8.

Applying Poisson Summation Formula to this function f we obtain

$$1 \geq \sum_{v \in E_8} f(v) = \sum_{w \in E_8} \hat{f}(w) \geq 1,$$

since the minimal length of the E_8 lattice is $\sqrt{2}$. Because $f(v) \leq 0$ for all $v \neq 0$ in E_8 and $\hat{f} \geq 0$, we conclude that $f(v) = \hat{f}(v) = 0$, for all $v \in E_8 \setminus \{0\}$. Moreover, the sign conditions also imply that the zeros in the points $\sqrt{2n}$ for $n \geq 2$ must be double for f and the zeros in $\sqrt{2n}$ for $n \geq 1$ must be double for $\hat{f}$.

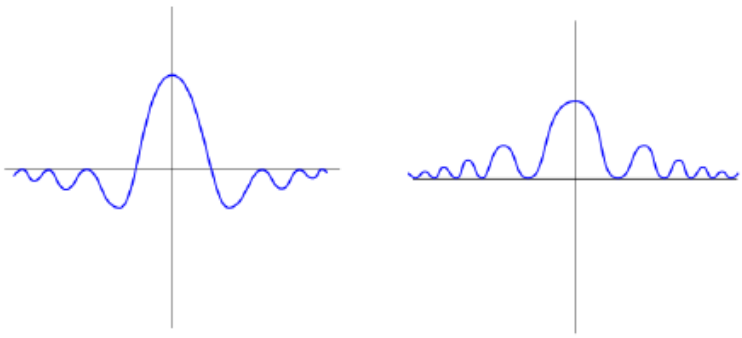


Figure 3: Sketch of the functions f and $\hat{f}$.

The same analysis can be done in dimension 24 for $r = 2$; the existence of such a function would prove that the Leech lattice packing is optimal. We obtain similar root conditions for f, $\hat{f}$ and their derivatives.

Also, if such optimal functions exist, we may assume that they are radial. Thus, if f is a optimal function than so $f \circ \rho$ is, for ρ a rotation. Taking the average over all the rotations, we obtain a rotationally symmetric optimal function. We recall that a radial function f can be written as the sum of a $+1$ and a -1-eigenfunction of the Fourier transform, by letting $f = a+b$, where $a = \frac{f+\hat{f}}{2}$ and $b = \frac{f-\hat{f}}{2}$. a and b also satisfy similar root conditions to f and $\hat{f}$: a has double zeros at $\sqrt{2n}$, for $n \geq 2$, and a simple zero at $\sqrt{2}$ and b has double zeros at $\sqrt{2n}$, for $n \geq 2$, and simple zeros at $\sqrt{2}$ and 0.

Instead of looking for an optimal function f satisfying the root conditions commented above, we impose more restrictions and look for eigenfunctions of the Fourier transform that have the roots that we seek. That is, we try to construct a and b and we expect that their sum satisfy the sign conditions needed.

One thing that is going to be useful in our search for a and b is the Cohn-Miller conjecture [5]. After running computations, they conjectured that the Taylor series of the magic function f in dimension 8 satisfy

$$\begin{aligned} f(x) &= 1 - \frac{27}{10}|x|^2 + O(|x|^4) \\ \hat{f}(x) &= 1 - \frac{3}{2}|x|^2 + O(|x|^4). \end{aligned} \tag{3.2}$$

4 Modular Forms

4.1 Motivation to the study of modular forms

We recall that we are looking for eigenfunctions of the Fourier transform satisfying some properties. One way of constructing eigenfunctions is considering the Laplace transform of modular forms. For instance, for $x \in \mathbb{R}^{2d}$, let

$$f(x) = \int_0^\infty g(it) e^{-\pi t|x|^2} dt,$$

where g is a function defined in the upper half-plane satisfying $g(i/t) = (it)^{2-d}g(it)$. We assume that g is well-behaved enough so that we can swap integrals and write

$$\hat{f}(x) = \int_0^\infty g(it)t^{-d}e^{-\pi|x|^2/t}dt.$$

Using the condition we asked on g, we conclude that

$$\hat{f}(x) = \int_0^\infty g(i/t)t^{d-2}e^{-\pi t|x|^2}dt = i^{2-d}\int_0^\infty g(it)e^{-\pi t|x|^2}dt = i^{2-d}f(x),$$

that is, f is a i^{2-d}-eigenfunction of the Fourier transform. If g is a modular form of weight $2-d$, it satisfies the required equality $g(i/t) = (it)^{2-d}g(it)$.

Considering f in that form will not solve our problem, since we have no control of the zeros of the function. However, an adaptation of this approach will gives us the eigenfunctions with the desired roots.

4.2 The full modular group

Let $\mathbb{H} = \{z \in \mathbb{C} : Re(z) > 0\}$ be the upper half-plane and Γ a subgroup of $SL(2,\mathbb{Z})$. Γ acts on $\mathbb{H}$ by the Möbius transformations:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} z = \frac{az+b}{cz+d}.$$

It is a simple computation to verify that composing two Möbius transformations is the same as multiplying the corresponding matrices. To check that this action actually takes $\mathbb{H}$ to $\mathbb{H}$, we write $z = u + iv$ and observe that

$$\begin{aligned} \mathrm{Im}\left(\frac{az+b}{cz+d}\right) &= \mathrm{Im}\left(\frac{a(u+iv)+b}{c(u+iv)+d}\right) \\ &= \mathrm{Im}\left(\frac{(ad-bc)v}{|cz+d|^2}\right) \\ &= \mathrm{Im}\left(\frac{v}{|cz+d|^2}\right) > 0. \end{aligned}$$

An important class of subgroups of $SL(2,\mathbb{Z})$ are

$$\Gamma(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \bmod N \right\}.$$

In our work, we are going to deal with $\Gamma(1)$, which is the full modular group $SL(2,\mathbb{Z})$, and with $\Gamma(2)$.

Before we focus our work in $\Gamma(1)$, we consider $\Gamma \subset \Gamma(1)$ acting in $\mathbb{H}$ and recall some general notions: two points $z_1, z_2 \in \mathbb{H}$ are **Γ-equivalent** if there exists a $\gamma \in \Gamma$ such that $\gamma z_1 = z_2$ and the **orbit** of a point z is given by $O_\Gamma(z) = \{\gamma z : \gamma \in \Gamma\}$. We say that $\mathcal{F} \subset \mathbb{H}$ is a **fundamental domain** for Γ if it is open, for all z_1, $z_2 \in \mathbb{H}$, $z_1 \notin O_\Gamma(z_2)$ and $\bar{\mathcal{F}} \cap O_\Gamma(z) = \emptyset$, for all $z \in \mathbb{H}$.

Theorem 4.1. *The region $\mathcal{F} = \{z \in \mathbb{H} : |z| > 1, |Re(z)| < 1/2\}$ is a fundamental domain for $\Gamma(1)$.*

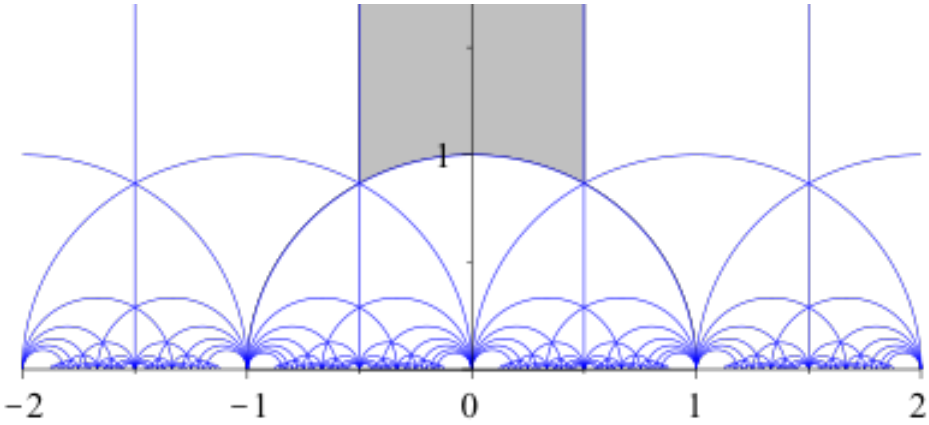


Figure 4: Fundamental domain for $\Gamma(1)$.

Proof. We first prove that if $z_1 \neq z_2 \in \mathcal{F}$, then $z_1 \notin O_\Gamma(z_2)$, that is, it cannot exist a $\gamma \in \Gamma(1)$ such that $z_2 = \gamma z_1$. Suppose, by contradiction, that such a $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ exits. Observe that c is different than zero, otherwise both a and d are equal to 1 or to -1 and this implies that γ is a translation by an integer. Since $|\mathrm{Re}(z_i)| < 1/2$, for $i = 1, 2$, this cannot hold.

Furthermore, since $\mathrm{Im}(z_i) > \sqrt{3}/2$, $i = 1, 2$, it holds that

$$\frac{\sqrt{3}}{2} < \mathrm{Im}(z_2) = \frac{\mathrm{Im}(z_1)}{|cz_1 + d|^2} \leq \frac{1}{c^2 \mathrm{Im}(z_1)} < \frac{2}{c^2\sqrt{3}},$$

which implies that $c = \pm 1$. To conclude this first statement, we suppose without loss of generality that $\mathrm{Im}(z_1) \leq \mathrm{Im}(z_2)$. Then,

$$\mathrm{Im}(z_2) = \frac{\mathrm{Im}(z_1)}{|\pm z_1 + d|^2} \leq \frac{\mathrm{Im}(z_1)}{|z_1|^2} < \mathrm{Im}(z_1),$$

since $|\pm z_1 + d| \geq |z_1| > 1$, giving us a contradiction.

Now let z be any point in $\mathbb{H}$. We want to show the existence of a $w \in \bar{\mathcal{F}}$ such that $w \in O_{\Gamma(1)}(z)$. The idea to find the desired w is to maximize the imaginary part of γz, for $\gamma \in \Gamma(1)$. We consider the lattice $\{mz + n : m, n \in \mathbb{Z}\}$ and let $cz + d$ be a non-zero point in this lattice that have minimal norm. One can observe that c and d must be coprime, otherwise $cz + d$ would not have minimal norm. Thus, there exists a and $b \in \mathbb{Z}$ such that $ad - bc = 1$, that is

$$\alpha = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(1).$$

Note that αz has the greatest imaginary part in the orbit of z since

$$\mathrm{Im}(\gamma z) = \frac{\mathrm{Im}(z)}{|c'z + d'|^2},$$

for all $\gamma = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \in \Gamma(1)$, and $cz + d$ minimizes the norm.

As the matrix $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, that takes z to $z+1$, belongs to $\Gamma(1)$, we know that there exists $k \in \mathbb{Z}$ such that $\mathrm{Re}(T^k(\alpha z)) \leq 1/2$. Let $w = T^k(\alpha z)$ and observe that to conclude the proof, it is enough to show that $|w| \geq 1$.

One can notice that we are only translating the real part of $\alpha(z)$, thus w also maximize the imaginary part among γz, $\gamma \in \Gamma(1)$. Observe that $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in \Gamma(1)$ and $Sw = -1/w$. If we suppose that $|w| < 1$, then

$$\mathrm{Im}(Sw) = \frac{\mathrm{Im}(w)}{|w|^2} > \mathrm{Im}(w),$$

which cannot happen, concluding the proof. □

A nice consequence of this theorem is that T and S, as defined above, together with $-I$ generate $\Gamma(1)$. In fact, one can observe that the neighbours of $\mathcal{F}$ are $T\mathcal{F}$, $T^{-1}\mathcal{F}$ and $S\mathcal{F}$. Then, if $\gamma \in \Gamma(1)$, the neighbours of $\gamma\mathcal{F}$ are given by moving this region by $\gamma T\gamma^{-1}$, $\gamma T^{-1}\gamma^{-1}$, $\gamma S\gamma^{-1}$. By induction, we can take $\mathcal{F}$ to $\gamma\mathcal{F}$ by applying T, T^{-1} are S repeatedly, which implies that any γ in $\Gamma(1)$ is a product of these matrices.

Corollary 4.2. *The full modular group* $\Gamma(1)$ *is generated by* S, T *and* $-I$.

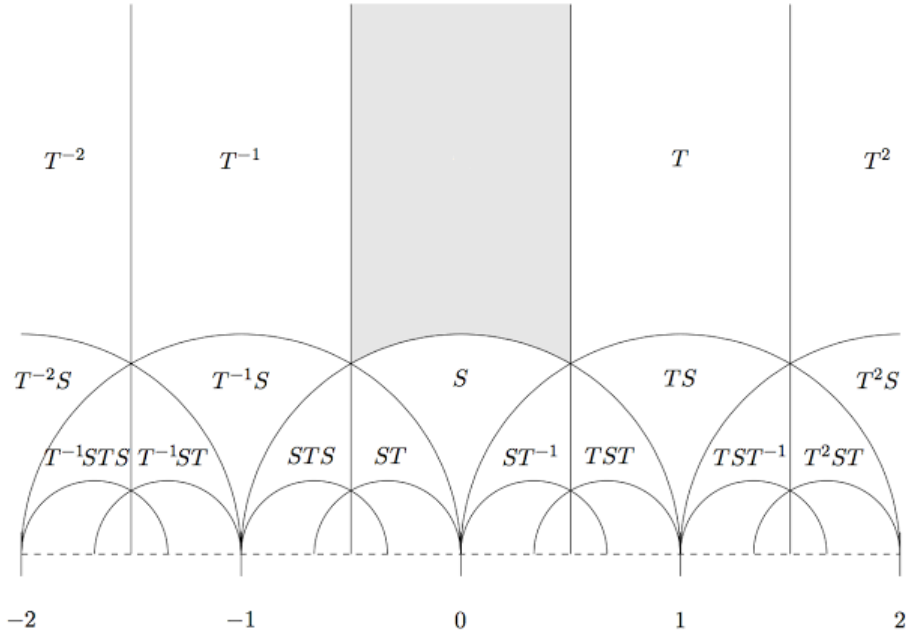


Figure 5: Tessellation of the hyperbolic plane.

4.3 Modular forms: generalities

Given any subgroup $\Gamma \subset SL(2,\mathbb{Z})$, we define a modular form of weight k over Γ as a holomorphic function $f : \mathbb{H} \to \mathbb{C}$ satisfying

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z), \tag{4.1}$$

for all $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$ and with a exponential decay

$$f(x+iy) = O(e^{Cy}) \text{ as } y \to \infty \textbf{ for all } C > 0$$

and

$$f(x+iy) = O(e^{C/y}) \text{ as } y \to 0 \textbf{ for all } C > 0.$$

We can weaken the decay conditions to

$$f(x+iy) = O(e^{Cy}) \text{ as } y \to \infty \textbf{ for a } C > 0$$

and

$$f(x+iy) = O(e^{C/y}) \text{ as } y \to 0 \textbf{ for a } C > 0,$$

and call f a weakly modular form. The implications of the decay conditions will be explored later. The set of modular forms of weight k over Γ is denoted by $M_k(\Gamma)$.

The **automorphy factor** of weight k is defined as

$$j_k\left(z, \begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = (cz+d)^{-k}.$$

If we let f be a function in $\mathbb{H}$ and $\gamma \in \Gamma$, the slash operator acts on f by

$$f|_k\gamma(z) = j_k(z,\gamma) f(\gamma z).$$

The modularity conditions in the definition of a modular form of weight k can be suppressed in the condition $f|_k\gamma = f$, for all $\gamma \in \Gamma$.

The first important property concerning modular forms that we observe is the chain rule that the slash operator satisfies, namely

$$j_k(z, \gamma\gamma') = j_k(z,\gamma') j_k(\gamma' z, \gamma). \tag{4.2}$$

This assertion can be proved by a straightforward computation.

We let $M_*(\Gamma) = \bigoplus_k M_k(\Gamma)$ and observe that it is a graded algebra. That is, the sum of modular forms of weight k is a modular form of weight k and the product of a modular form of weight k and a modular form of weight l is a modular form of weight $k+l$. The weight of a non-zero modular form is always even. Indeed, if f is a modular form of weight k, then $f(z) = f((-I)z) = (-1)^k f(z)$, which only holds for odd k if f is identically zero.

The most important and useful properties of modular forms are the modularity conditions and their Fourier expansion (see some applications in [1, Chap. 1]). This last property derives from our hypothesis on the growth of the function.

Theorem 4.3. *Let $\Gamma \subset \Gamma(1)$ be a subgroup that contains T^N, for some $N \in \Gamma$. Then, for any $f \in M_k(\Gamma)$ there exists c_j such that $f(z) = \sum_{j=0}^{\infty} c_j q^{j/N}$, for all $z \in \mathbb{H}$, where $q = e^{2\pi i z}$.*

Proof. We first observe that $T^N \in \Gamma$ implies that f is N-periodic. Then, $\eta_y(x) = f(Nx + iy)$ is 1-periodic and it is differentiable, which implies that it can be expanded in Fourier series. That is,

$$\eta_y(x) = \sum_{j \in \mathbb{Z}} c_j(y) e^{2\pi i j x},$$

where

$$c_j(y) = \int_0^1 f(Nx + iy) e^{-2\pi i x j} dx.$$

We can write f evaluated in $z = x + iy$ as

$$\begin{aligned} f(x + iy) = \eta_y\left(\frac{x}{N}\right) &= \sum_{j \in \mathbb{Z}} c_j(y) e^{2\pi i j x/N} \\ &= \sum_{j \in \mathbb{Z}} c_j(y) e^{2\pi i j x/N} e^{-2\pi j y/N} e^{2\pi j y/N} \\ &= \sum_{j \in \mathbb{Z}} c_j(y) e^{2\pi y/N} q^{j/N}. \end{aligned}$$

To conclude the proof, we show that $c_j(y) e^{2\pi y/N}$ is constant in y for each j and it is zero for $j < 0$. To show that the coefficients of the Fourier series are constant in y, we observe that

$$\begin{aligned} c_j(y) e^{2\pi y/N} &= e^{2\pi y/N} \int_0^1 f(Nx + iy) e^{-2\pi i x j} dx \\ &= e^{2\pi y/N} e^{-2\pi y/N} \frac{1}{N} \int_{iy}^{iy+N} f(w) e^{-2\pi i j w/N} dw = \frac{1}{N} \int_{iy}^{iy+N} f(w) e^{-2\pi i j w/N} dw, \end{aligned}$$

where we made the change of variables $w = Nx + iy$, and the last two integrals are over the line that connects iy and $iy + N$. The function $f(w) e^{-2\pi i j w/N}$ is holomorphic and has period N, which implies that derivative of $\int_{iy}^{iy+N} f(w) e^{-2\pi i j w/N}$ is equal to zero, concluding the first statement.

Finally, we recall that for all $C > 0$, $|f(x + iy)| \leq c(x) e^{Cy}$, where $c(x)$ is a constant that depends on x. Let $C < 2\pi/N$ and $c > 0$ be such that $|f(x + iy)| \leq c e^{Cy}$, as $y \to \infty$, for all $0 \leq x \leq N$, then for $j < 0$

$$|c_j e^{2\pi y/N}| = \left| e^{2\pi y/N} \int_0^1 f(Nx + iy) e^{-2\pi i x j} dx \right| \leq e^{2\pi j y/N} \frac{1}{N} \int_0^N |f(t + iy)| dt \leq c e^{(C + 2\pi j/N) y},$$

which goes to zero as $y \to \infty$. Thus, $c_j = 0$ for all $j < 0$.

□

One can observe that a similar result holds for weakly modular forms as well. If we assume that there exists a $C > 0$ such that $|f(x + iy)| \lesssim e^{Cy}$, as $y \to \infty$, the last part of proof does not

hold necessarily for every $j < 0$ but it does for every $j < j_0$ that depends on the C. Therefore, we conclude that if f is a weakly modular form over Γ, there exists a j_0 such that

$$f(z) = \sum_{j \geq j_0} c_j q^{j/N}.$$

4.4 Modular forms over $\Gamma(1)$

We now focus our attention to the construction of modular forms over $\Gamma(1)$ and explore some properties that $M_*(\Gamma(1))$ must satisfy.

As observed in corollary 4.2, the full modular group is finitely generated by T, S and $-I$. Together with the chain rule 4.2 that holds for the slash operator, we note that to check that a function satisfies the modulating properties in equation 4.1 it is sufficient to look at the generators. That is, if $f : \mathbb{H} \to \mathbb{C}$ is such that

$$\begin{aligned} f(z+1) &= f(z) \\ f(-1/z) &= z^k f(z), \end{aligned} \tag{4.3}$$

then for every $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(1)$,

$$f\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} z\right) = (cz+d)^k f(z).$$

Among well-known examples of modular forms over $\Gamma(1)$ we have the Eisenstein series. For even $k > 2$, we define

$$E_k(z) = \frac{1}{\zeta(k)} \sum_{(m,n) \neq (0,0) \in \mathbb{Z}^2} \frac{1}{(mz+n)^k}.$$

It is not hard to see that E_k is a modular form of weight k by checking the properties 4.3; they follow by rearranging the sums.

The Fourier expansions of the Eisenstein series are interesting by themselves since they immediately imply non-trivial identities in number theory, see more in [1, Chap. 2].

Proposition 4.4. *The Fourier expansion of E_k, for $k > 2$, is*

$$E_k(z) = \frac{1}{\zeta(k)} \left(-\frac{B_k}{2k} + \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n \right),$$

where B_k is the kth Bernoulli number, $\sigma_{k-1}(n)$, for $n \in \mathbb{N}$, denotes the sum of the $(k-1)$ powers of the positive divisors of n, and $q = e^{2\pi i z}$.

The proof of the result can be found in [1, pg. 16]. One can compute the first terms of the expansion of some Eisenstein series:

$$\begin{aligned} E_4(z) &= 1 + 240q + 2160q^2 + \cdots \\ E_6(z) &= 1 - 504q - 16632q^2 + \cdots \\ E_8(z) &= 1 + 480q + 61920q^2 + \cdots . \end{aligned} \tag{4.4}$$

Another important feature of the Eisenstein series is the fact that they generate the algebra $M_*(\Gamma(1))$. We will not prove it, but one can find in [1, pg. 15] that the following holds:

Theorem 4.5. *If $k > 0$ is an even integer then*

$$dim M_k(\Gamma(1)) = \begin{cases} \lfloor \frac{k}{12} \rfloor + 1 & \text{if } x \not\equiv 2 \ mod\ 12 \\ \lfloor \frac{k}{12} \rfloor & \text{if } x \equiv 2 \ mod\ 12. \end{cases}$$

Moreover, the ring $M_*(\Gamma(1))$ is freely generated by the modular forms E_4 and E_6.

In other words, the set $\{E_4^j E_6^i : i, j \geq 0; 4i + 6j = k\}$ is a basis for $M_k(\Gamma(1))$. This property will be very important for the construction of the magic functions.

One can observe in the previous theorem that $k = 12$ is the first integer such that $\dim M_k(\Gamma(1)) > 1$. In fact, $\dim M_{12}(\Gamma(1)) = 2$ and this space is given by linear combinations of E_6^2 and E_4^3. We call the modular form of weight 12 that has linear term equal to 0 and coefficient of the q term equal to 1 the discriminant function Δ. From equation 4.4, one can see that $\Delta = \frac{E_4^3 - E_6^2}{1728}$. Another way of writing the discriminant function ([1, pg. 21]) is the following

$$\Delta(z) = q \prod_{n=1}^{\infty} (1 - q^n)^{24},$$

for $q = e^{2\pi i z}$. In this form, it is easy to see that this function never vanishes in the upper half -plane. This property will be important to the construction of weakly modular forms of weight smaller than 4, since we can divide modular forms by Δ without creating poles.

4.5 Quasimodular forms

One can see from theorem 4.5 that there are no modular forms of weight 2. If we let $k = 2$ in the definition of the Eisenstein series, we can arrange the sum in order to have a well-defined function, namely

$$E_2(z) = \frac{1}{\zeta(2)} \left(\sum_{m \in \mathbb{Z} \setminus \{0\}} \sum_{n \in \mathbb{Z}} \frac{1}{(mz + n)^2} + \sum_{n \in \mathbb{Z}} \frac{1}{n^2} \right).$$

The restraint that do not allow E_2 be a modular form is the modulating property $f(-1/z) = z^2 f(z)$. One can prove that E_2 has period one, exactly the same way it is proven that this property holds for the Eisenstein series. However, if we try to prove the second modulating property as it is done with E_k, for $k > 2$, we realize that the series need to be exchanged but it is not possible, since they do not converge absolutely.

However, E_2 defines what we call a quasimodular form, because it holds that $E_2(-1/z) = z^2 E_2(z) - 6iz/\pi$ rather than $E_2(-1/z) = z^2 E_2(z)$ - the proof can be found in [1, pg. 19]. This extra term that prevent E_2 from being a modular form will be indispensable for the construction of the magic functions.

We can define, more generally, the space of quasimodular of weight k and depth d over $\Gamma(1)$ as $\widetilde{M}_k^{\leq d}(\Gamma(1)) = \{\sum_{j=0}^{d} f_j E_2^j : f_j \in M_{k-2j}(\Gamma(1))\}$. Analogously, we define the space of weakly quasimodular forms by considering the f_j weakly modular forms.

4.6 Theta series

Considering modular forms over $\Gamma(1)$ is not going to be sufficient to construct the magic functions and we also need to take a look at the modular forms over $\Gamma(2)$. We introduce some concepts of the theory of *theta series*. These objects are closely related to modular forms and lattices and have remarkable properties that lead to many applications in number theory and combinatorics. The proofs of the results stated bellow such as some applications of the theta series theory can be found in [1, Chap. 3]. Here we follow the notation of [12] to define them.

The theta series that are useful to us are defined as follows:

$$\begin{aligned}\theta_{00}(z) &= \sum_{n\in\mathbb{Z}} e^{\pi i n^2 z} \\ \theta_{01}(z) &= \sum_{n\in\mathbb{Z}} (-1)^n e^{\pi i n^2 z} \\ \theta_{10}(z) &= \sum_{n\in\mathbb{Z}+\frac{1}{2}} e^{\pi i n^2 z}.\end{aligned} \tag{4.5}$$

If we take the 4th power of these series, we obtain modular forms of weight 2 on $\Gamma(2)$. Analysing their q-expansion one can observe that

$$\begin{aligned}\theta_{00}^4(z+1) &= \theta_{01}^4(z) \\ \theta_{01}^4(z+1) &= \theta_{00}^4(z) \\ \theta_{10}^4(z+1) &= -\theta_{10}^4(z),\end{aligned}$$

and Poisson Summation formula yields

$$\begin{aligned}z^{-2}\theta_{00}^4(-1/z) &= -\theta_{00}^4(z) \\ z^{-2}\theta_{01}^4(-1/z) &= -\theta_{10}^4(z) \\ z^{-2}\theta_{10}^4(-1/z) &= -\theta_{01}^4(z).\end{aligned}$$

Another important property is the Jacobi identity:

$$\theta_{00}^4(z) = \theta_{01}^4(z) + \theta_{10}^4(z). \tag{4.6}$$

Together with these relations, an important fact about these theta series is the following

Theorem 4.6. *The space of modular forms of* $\Gamma(2)$ *is freely generated by any pair of the triplet* θ_{00}^4, θ_{10}^4 *and* θ_{01}^4.

5 An overview of Viazovska's proof

By now we have all the necessary tools to analyse Viazovska's proof on the sphere packing problem. We work backwards on the proof and try to deduce how the magic functions of Cohn and Elkies theorem look like considering their properties. The ideas of the constructions in dimensions 8 and 24 are similar, thus we focus in dimension 8 for simplicity.

Recall that we want to construct a radial Schwartz function $f : \mathbb{R}^8 \to \mathbb{R}$ such that $f(0) = \hat{f}(0) = 1$, $\hat{f}(x) \geq 0$ for all $x \in \mathbb{R}^8$ and $f(x) \leq 0$ for all $|x| \geq \sqrt{2}$. In Section 3, we deduced that if such a f exists, it can be written as $f = a + b$, where a and b are $+1$ and -1 eigenfunctions of the Fourier transform, respectively. Also, a and b must have double zeros at $\sqrt{2n}$, for $n \geq 2$ and a simple zero at $\sqrt{2}$, and b also has a extra simple zero at 0.

In Section 4, we observed that modular forms can originate eigenfunctions of the Fourier transform. If ψ is a well-behaved enough weakly modular form of weight -2 then

$$g(x) = \int_0^\infty \psi(it) e^{-\pi t|x|^2} dt,$$

is a -1-eigenfunction in $\mathbb{R}^8$.

The problem with this definition is that we have no control of the zeros of the function. Viazovska's great idea was to multiply the expression above by a sine factor that produces the desired roots, namely,

$$g(x) = \sin^2\left(\pi|x|^2/2\right) \int_0^\infty \psi(it) e^{-\pi t|x|^2} dt.$$

One can observe that such an expression could potentially have more zeros than needed. For instance, $\sin^2\left(\pi|x|^2/2\right)$ has a root of order 4 at zero and double roots at $\pm\sqrt{2}$. We investigate what are the modulating properties and the decay conditions that ψ has to satisfy in order to give rise to eigenfunctions of the Fourier transform with the exact roots.

5.1 The $+1$-eigenfunction

We set

$$a(r) = \sin^2(\pi r^2/2) \int_0^\infty \psi(it) e^{-t\pi r^2} dt,$$

where we $r = |x|^2$, and try to find the hypothesis on ψ so that a satisfies the conditions stated above. Expanding the sine factor in exponentials we obtain

$$\begin{aligned} -4a(r) &= \int_0^\infty \psi(it) e^{-(t+i)\pi r^2} dt - 2\int_0^\infty \psi(it) e^{-\pi r^2 t} dt + \int_0^\infty \psi(it) e^{-(t-i)\pi r^2} dt \\ &= \frac{1}{i}\left(\int_{-1}^{-1+i\infty} \psi(s+1) e^{i\pi r^2 s} ds - 2\int_0^{i\infty} \psi(s) e^{i\pi r^2 s} dt + \int_1^{1+i\infty} \psi(s-1) e^{i\pi r^2 s} ds \right). \end{aligned} \tag{5.1}$$

If we assume that ψ is well-behaved enough in order to take the Fourier transform inside of the integral, we obtain:

$$\begin{aligned} -4i\hat{a}(r) = &\int_{-1}^{-1+i\infty} \psi(z+1) z^{-4} e^{i\pi r^2(-1/z)} dz - 2\int_0^{i\infty} \psi(z) z^{-4} e^{i\pi r^2(-1/z)} dz \\ &+ \int_1^{1+i\infty} \psi(z-1) z^{-4} e^{i\pi r^2(-1/z)} dz. \end{aligned} \tag{5.2}$$

We seek the equality $a(r) = \hat{a}(r)$, hence we want to be able to compare the expressions 5.1 and 5.2 and obtain identities on ψ. If the contours of the integrals and the exponentials inside

the integrals were the same we could do it. Aiming that, we manipulate the expression 5.2 to get one similar to 5.1.

Our first attempt is making the change of variables $t \mapsto -1/t$. We get

$$\begin{aligned} -4i\hat{a}(r) = &\int_C \psi(-1/z+1)t^2 e^{i\pi r^2 z} dz + 2\int_0^{i\infty} \psi(-1/z) z^2 e^{i\pi r^2 z} dz \\ &+ \int_D \psi(-1/z-1) z^2 e^{i\pi r^2 z} dz, \end{aligned} \tag{5.3}$$

where C and D are the contours illustrated bellow.

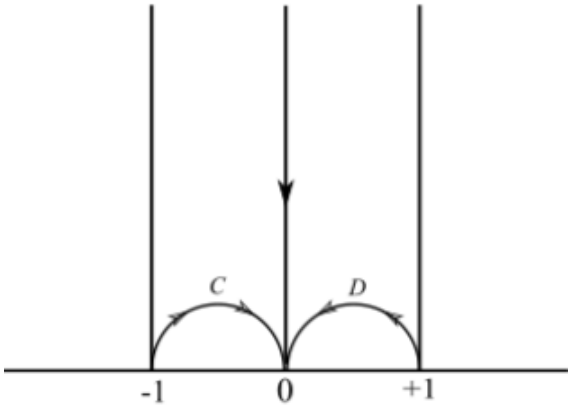


Figure 6: Contour integration.

Here we come across the following problem: we want to deform the set of contours in 5.1 in such a way that when we change variables $t \mapsto -1/t$ in the expression for the Fourier transform the contours in this new set interchange between each other and we do not create a different path. The line that goes from 0 to $i\infty$ just change the direction, therefore we leave it the same. The idea for the remaining paths is the following: instead of integrating in the line that goes from -1 to $-1+i\infty$, we integrate from -1 to i along the circle of radius 1 and then in the line that connects i and $i\infty$. Analogously, we integrate from 1 to and i along the circle and from i to $i\infty$ along the line. One can see that if $t \mapsto -1/t$, the semicircle that starts in -1 and ends in i oriented clockwise becomes the semicircle that starts in 1 and ends in i oriented counter clockwise, and vice-versa. The line that goes from i to $i\infty$ becomes the line from 0 to i, which is considered in the second contour of 5.1.

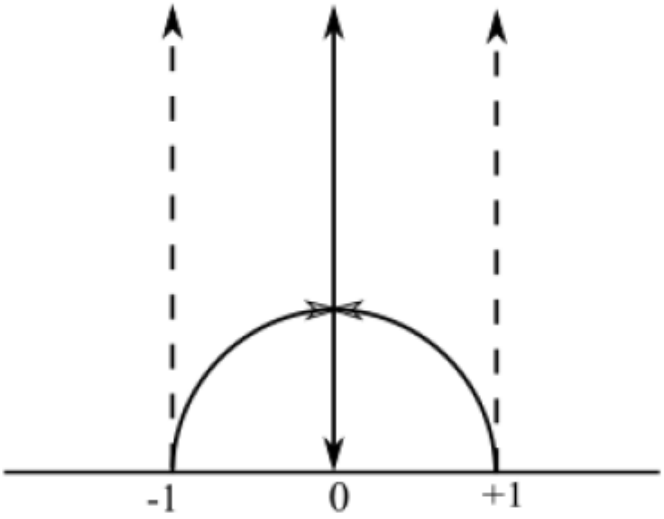


Figure 7: New contour integration.

Of course, to be able to do this shift of contours, the function ψ must have a good decay and cannot have poles between the region above the unit circle and in the lines of real part $+1$ and -1 in $\mathbb{H}$. We assume that ψ satisfies these properties and proceed with the analysis.

The contour shift yields

$$\begin{aligned} -4ia(r) = &\int_i^{i\infty} (\psi(z+1) - 2\psi(z) + \psi(z-1))e^{i\pi r^2 z}dz - 2\int_0^i \psi(z)e^{i\pi r^2 z}dz \\ &+ \int_{-1}^i \psi(z+1)e^{i\pi r^2 z}dt + \int_1^i \psi(z-1)e^{i\pi r^2 z}dz \end{aligned} \tag{5.4}$$

and

$$\begin{aligned} -4i\hat{a}(r) = &\int_i^0 (\psi(-1/z+1) - 2\psi(-1/z) + \psi(-1/z-1))z^2 e^{i\pi r^2 z}dz + 2\int_i^{i\infty} \psi(-1/z)z^2 e^{i\pi r^2 z}dz \\ &+ \int_1^i \psi(-1/z+1)z^2 e^{i\pi r^2 z}dz + \int_{-1}^i \psi(-1/z-1)z^2 e^{i\pi r^2 z}dz. \end{aligned} \tag{5.5}$$

If we equate the integrands in the expressions 5.4 and 5.5 we obtain the identities:

$$\psi(z+1) - 2\psi(z) + \psi(z-1) = 2\psi(-1/z)z^2 \tag{5.6}$$

$$2\psi(z) = (\psi(-1/z+1) - 2\psi(-1/z) + \psi(-1/z-1))z^2 \tag{5.7}$$

$$\psi(z+1) = \psi(-1/z-1)z^2 \tag{5.8}$$

$$\psi(z-1) = \psi(-1/z+1)z^2. \tag{5.9}$$

By letting $z = -1/s$ in expressions 5.6 and 5.8 we realize that identity 5.6 is equivalent to 5.7 and that identity 5.8 is equivalent to 5.9. We study more closely equalities 5.6 and 5.8. If we let $z + 1 = -1/s$ in expression 5.8 we get that

$$\psi\left(-\frac{1}{s}\right)s^2 = \psi\left(-\frac{1}{s+1}\right)(s+1)^2, \tag{5.10}$$

that is, $\phi(z) := \psi(-1/z)z^2$ is a periodic function of period 1. Therefore, we expect ϕ to have a q-expansion and we hope to construct it out of modular forms. However, we realize that modular forms will not solve the problem. We recall that the sine factor in the definition of a has a zero of order 4 at 0 and that we want $a(0) = \frac{f(0)+\hat{f}(0)}{2} = 1$. Thus, we need the Laplace transform of ψ to have a pole of order 4 at 0.

If we assume that ϕ is a weakly modular form, that is, $\phi = Q(E_4, E_6)$ for Q a rational function, then $\psi(it) = -t^2Q(E_4, E_6)(i/t)$. Using the modularity properties, we see that this last expression can be written as a polynomial of even powers of t times a q-expansion. The only way to create a pole of order 4 through the Laplace transform of a polynomial times exponentials is having a 'naked' term t, since

$$\int_0^\infty te^{-\pi r^2 t}dt = \frac{1}{\pi^2 r^4}.$$

But we cannot generate an odd power of t if ϕ is a weakly modular form, as observed above. Therefore, we let the non-modular E_2 play the game: recall that $E_2(i/t) = -E_2(it) + 6t/\pi$, which has a extra term t. Hence, we assume from now on that ϕ is a weakly quasimodular form.

We turn ourselves to expression 5.6. It can be expressed in terms of ϕ as

$$\psi(z+1) - 2\psi(z) + \psi(z-1) = 2\phi(z), \tag{5.11}$$

or alternatively

$$\Delta^{(2)}(z^2\phi(-1/z)) = 2\phi(z). \tag{5.12}$$

We observe that if ϕ is a weakly quasimodular form then it can be written as $\phi = Q(E_2, E_4, E_6)$, for Q a rational function, and it follows that $t^2\phi(i/t) = t^k P(E_2, E_4, E_6)(it)$, for some k and a rational function P. Taking the second finite difference we get $\Delta^{(2)}(t^k P(E_2, E_4, E_6)(it)) = \Delta^{(2)}(t^k)P(E_2, E_4, E_6)(it)$, by the periodicity of the Eisenstein series. Thus, to equality 5.12 be true, we must have

$$\phi(-1/z)z^2 = z^2\phi(z) + z\phi_1(z) + \phi_2(z), \tag{5.13}$$

where ϕ_i, $i = 1, 2$, are weakly quasimodular forms.

The remaining tasks to do are determining the weight and depth of ϕ. From identity 5.13 we have that

$$\phi(-1/z) = \phi(z) + z^{-1}\phi_1(z) + z^{-2}\phi_2(z), \tag{5.14}$$

which implies that ϕ must have weight 0.

Now we recall that the discriminant function Δ has no zeros in upper half-plane. Thus, to create weakly modular forms of weight 0, we consider functions of the form

$$\frac{P(E_2, E_4, E_6)}{\Delta^n},$$

where $P(E_2, E_4, E_6)$ is a quasimodular form of weight $12n$. One can notice that $n = 1$. In fact, expanding the numerator we get $t^2 P(E_2, E_4, E_6)(i/t) = p(t)(c_0 + c_1 e^{-2\pi t} + c_2 e^{-4\pi t} + \cdots)$ and from the denominator we get $\Delta^n(i/t) = t^{12n}(e^{-2n\pi t} + \cdots)$. We know that for $r^2 > 2$ the Laplace transform must be holomorphic, since it already has the right roots of the right orders. Then, for $r^2 > 2$, the term $c_0 p(t) t^{-12n} e^{-(r^2-2n)\pi t}$ must be integrable at ∞, which is only possible if $n = 1$.

Finally, we check that the depth of ϕ must be at most 2. Suppose that ϕ has a factor $E_2^\alpha E_4^\beta E_6^\gamma$, where $2\alpha + 4\beta + 6\gamma = 12$. Then, we are going to integrate the following expression from 0 to ∞, and for $r^2 > 2$:

$$\begin{aligned}
&\frac{t^2 E_2^\alpha(i/t) E_4^\beta(i/t) E_6^\gamma(i/t)}{\Delta(i/t)} e^{-\pi r^2 t} = \frac{t^2(-t^2 E_2(it) + \frac{6}{\pi} t)^\alpha t^{4\beta} E_4(it)(it)^{6\gamma} E_6(it)}{t^{12}\Delta(it)} e^{-\pi r^2 t} \\
&= -\frac{(\frac{6}{\pi})^\alpha t^2 t^\alpha t^{4\beta} t^{6\gamma}}{t^{12}} \frac{E_4(it) E_6(it)}{\Delta(it)} e^{-\pi r^2 t} + (\text{terms in t of higher order times q-expansion}) e^{-\pi r^2 t}.
\end{aligned} \tag{5.15}$$

We already know that exponential factors are integrable at infinity, by the observation we made before. To guarantee that expression 5.15 is integrable at zero, the lower order of the t factor must be at least 0, that is

$$\alpha + 4\beta + 6\gamma \geq 10.$$

Summing α in both sides of the inequality, we conclude that

$$12 \geq 10 + \alpha \Rightarrow \alpha \leq 2,$$

as we wanted. In summary, ϕ is a linear combination of E_4^3, E_6^2, $E_2 E_6 E_4$ and $E_2^2 E_4^2$, divided by the discriminant function:

$$\phi = \frac{A E_4^3 + B E_6^2 + C E_2 E_6 E_4 + D E_2^2 E_4^2}{\Delta}. \tag{5.16}$$

To determine the constants A, B, C and D we use the information about the roots of the function a. Observe that

$$\phi(-1/z) = \phi(z) + z^{-1} \frac{-\frac{6i}{\pi} C E_4 E_6 - \frac{12i}{\pi} D E_4^2 E_2}{\Delta} + z^{-2} \frac{\frac{-36}{\pi^2} D E_4^2}{\Delta}. \tag{5.17}$$

From equality 5.14 we conclude that

$$\begin{aligned}
\phi_1 &= \frac{-\frac{6i}{\pi} C E_4 E_6 - \frac{12i}{\pi} D E_4^2 E_2}{\Delta} \\
\phi_2 &= \frac{\frac{-36}{\pi^2} D E_4^2}{\Delta}.
\end{aligned}$$

To compute the constants C and D, we expand in Taylor the sine factor and the Laplace transform. Together with the information about the value of a in 0 and the root in $\pm\sqrt{2}$ it follows that

$$C = -\frac{\pi}{1080}, \quad D = \frac{\pi}{2160}.$$

To solve for A and B we observe that ϕ cannot have neither a q^{-1} nor a constant term, otherwise we would create undesired poles. These two informations gives us that

$$A = 0, \quad B = \frac{\pi}{2160}.$$

Therefore, we conclude that

$$\phi = \frac{\pi}{2160}\frac{(E_2E_4 - E_6)^2}{\Delta}.$$

5.2 The -1-eigenfunction

To construct the -1-eigenfunction, we start the same way we did with the $+1$-eigenfunction: consider

$$b(r) = \sin^2(\pi r^2/2)\int_0^\infty g(it)e^{-t\pi r^2}dt,$$

and make the same contour shifting as before. Comparing the expressions for b and $\hat{b}$ to get the equality $b = -\hat{b}$, we again obtain two identities:

$$g(z+1) - 2g(z) + g(z-1) = -2g(-1/z)z^2 \tag{5.18}$$

$$g(z+1) = -g(-1/z-1)z^2. \tag{5.19}$$

The equality 5.19 yields

$$z^2g(1/z) = -(z+1)^2g(-1/(z+1)),$$

that is, the function $\phi(z) = z^2g(1/z)$ satisfies

$$\phi(z+1) = -\phi(z). \tag{5.20}$$

We observe that equality 5.20 tells us that we can expect ϕ to have a q-expansion with only odd powers of $q^{1/2}$.

Expression 5.18 can be written as $\Delta^{(2)}(z^2\phi(-1/z)) = -2\phi(z)$. Observe that if we let g be a weakly modular form of weight -2 over $\Gamma(2)$ satisfying $z^2g(-1/z) + g(z+1) = g(z)$, then condition 5.18 holds. Indeed,

$$g(z-1) - 2g(z) + g(z+1) = 2(g(z+1) - g(z)) = -2z^2g(-1/z) = -2\phi(z).$$

Recall that for an even $k \geq 2$, it holds that $M_k(\Gamma(2)) = \text{spam}\{(\theta_{00}^4)^j(\theta_{01}^4)^{k/2-j} : j = 0, ..., k/2\}$. Then, we write

$$g = \frac{P(\theta_{00}^4, \theta_{01}^4)}{\Delta},$$

where P is an homogeneous polynomial of degree 5. This gives us

$$g = \frac{A\theta_{00}^{20} + B\theta_{00}^{16}\theta_{01}^4 + C\theta_{00}^{12}\theta_{01}^8 + D\theta_{00}^8\theta_{01}^{12} + E\theta_{00}^4\theta_{01}^{16} + F\theta_{01}^{20}}{\Delta},$$

and, as we did before, we use the constraints we know about b to determine the constants A, B, ..., F.

The equality $\phi(z+1) = -\phi(z)$ and Jacobi identity yields:

$$A = 0, \quad C = -2B, \quad E = B - D.$$

The expression 5.18 gives an extra constraint $B = 2D - 5F$. Observe that this implies that we only have two free parameters, since

$$A = 0, \quad B = 2D - 5F, \quad C = -4D + 10F, \quad E = D - 5F.$$

One can notice that a factor $q^{1/2}$ in g expansion would create a pole in ± 1, since

$$\int_0^\infty e^{\pi t} e^{-\pi r^2 t} dt = \frac{1}{\pi(r^2 - 1)}.$$

Thus, we set the coefficient of $q^{1/2}$ equal to zero, giving us that $5F = 2D$ and we only have one free parameter. To set the value of D we expand the sine in Taylor series and use Cohn-Miller conjecture 3.2. This gives us

$$g = -\frac{1}{120\pi} \frac{5\theta_{00}^8\theta_{01}^{12} - 5\theta_{00}^4\theta_{01}^1 6 + \theta_{01}^{20}}{\Delta}. \tag{5.21}$$

6 Formal proof

6.1 Verifying previous steps

In the last section, we found our candidates for the eigenfunctions of the Fourier transform. During our search for them, we made some manipulations that need to be justified. For instance, we swapped the Fourier transform and the Laplace transform integrals and we shifted the contours of integration. To show that $f = a + b$ is indeed the magic function, we also need to check that it satisfies the sign conditions of Cohn and Elkies theorem and behave correctly at 0 and $\pm\sqrt{2}$. The sign verifications need computational assistance to be verified and will only be commented.

We set

$$\phi_+ = \frac{\pi}{2160} \frac{(E_2E_4 - E_6)^2}{\Delta}, \tag{6.1}$$

and make some observations of its Fourier expansion. It holds that

$$\phi_+ = 240\pi(q + 60q^2 + 1680q^3 + 30280q^4 + \dots). \tag{6.2}$$

Using the modularity properties of the Eiseistein series, we also have that

$$\phi_+(-1/z)z^2 = \phi_+(z)z^2 + \frac{1}{180}\frac{E_4E_6 - E_4^2E_2}{\Delta}z - \frac{1}{60\pi}\frac{E_4^2}{\Delta} =: \phi_+(z)z^2 + \phi_1(z)z + \phi_2(z). \tag{6.3}$$

We denote $\phi_+(-1/z)z^2 = g_+(z)$ and observe that

$$\begin{aligned} g_+(z) &= \pi(240q + 14400q^2 + 403200q^3 + 7267200q^4 + \cdots)z^2 \\ &+ (-4i - 1128iq - 52320iq^2 - \dots)z + \frac{1}{\pi}\left(-\frac{q^{-1}}{60} - \frac{42}{5} - \frac{6147q}{5} - \frac{134752q^2}{3} - \dots\right). \end{aligned} \tag{6.4}$$

Evaluating in $z = it$ we get

$$-\phi_+(i/t)t^2 =: -\phi_+(it)t^2 + \phi_1(it)it + \phi_2(it). \tag{6.5}$$

To justify our manipulations, we need to obtain a bound on the Fourier coefficients of $\phi_+(-1/z)z^2$. We let $c_f(n)$ be the n-th Fourier coefficient of f, for $f = \phi_+, \phi_1, \phi_2$.

We will not prove it, but one can find in [11, pg. 67] that the following holds

Proposition 6.1. *For $n \geq 6$,*

$$\begin{aligned} c_{\phi_+}(n) &\leq \frac{1}{30\pi}e^{4\pi\sqrt{n}}, \\ |c_{\phi_1}(n)| &\leq \frac{1}{30\pi}e^{4\pi\sqrt{n}}, \\ |c_{\phi_2}(n)| &\leq \frac{1}{30\pi}e^{4\pi\sqrt{n}}. \end{aligned}$$

Observe that this implies that

$$|\phi_+(z)| \leq C_1\sum_{n=1}^{\infty} e^{4\pi\sqrt{n}}|e^{-2\pi nz}| \leq Ce^{-2\pi Im(z)}, \tag{6.6}$$

if $\mathrm{Im}(z) \geq 1/2$.

In view of expression 5.4 we define, for all $x \in \mathbb{R}^8$, $|x| = r$,

$$a(r) = -\frac{1}{4i}\left(\int_{-1}^{i} g_+(z+1)e^{\pi izr^2}dz + \int_1^i g_+(z-1)e^{\pi izr^2}dz - 2\int_0^i g_+(z)e^{\pi izr^2}dz + 2\int_i^{i\infty}\phi_+(z)e^{\pi izr^2}dz\right), \tag{6.7}$$

and observe that it is a Schwartz function. We first bound the last integral of 6.7: by 6.6 we get that

$$\left|\int_i^{i\infty}\phi_+(z)e^{\pi izr^2}dz\right| \leq \int_1^{\infty}|\phi_+(it)|e^{-\pi tr^2}dt \leq C\int_1^{\infty}e^{-2\pi t}e^{-\pi tr^2}dt = C\frac{e^{-\pi(r^2+2)}}{\pi(2+r^2)},$$

which decays faster than any polynomial. It is not hard to see that something similar holds for the derivatives of the Laplace transform.

We analyse now the first integral of 6.7 - the second and the third follows by analogous arguments. Using 6.6 again, we have

$$\begin{aligned}\left|\int_{-1}^{i}\phi(-1/(z+1))(z+1)^2e^{\pi ir^2z}dz\right| &= \left|\int_{(i-1)/2}^{i\infty-1/2}\phi(s)s^{-4}e^{\pi ir^2(-1/s-1)}ds\right| \\ &\leq C\int_{1/2}^{\infty}e^{-2\pi t}e^{-\pi r^2/t}dt \leq C\int_{0}^{\infty}e^{-2\pi t}e^{-\pi r^2/t}dt.\end{aligned}$$

For $r>0$, we split the integral

$$\int_0^\infty e^{-2\pi t}e^{-\pi r^2/t}dt = \int_0^{r/\sqrt{2}} e^{-2\pi t}e^{-\pi r^2/t}dt + \int_{r/\sqrt{2}}^\infty e^{-2\pi t}e^{-\pi r^2/t}dt$$

and analyse each of them separately. We bound the first one as follows

$$\int_0^{r/\sqrt{2}} e^{-\pi r^2/t}dt = \int_{\sqrt{2}/r}^\infty e^{-\pi sr^2}s^{-2}ds \leq \frac{r^2}{2}\int_{\sqrt{2}/r}^\infty e^{-\pi sr^2}ds \leq \frac{e^{-\pi\sqrt{2}r}}{2\pi}.$$

For the second one, we observe that

$$\int_{r/\sqrt{2}}^\infty e^{-2\pi t}e^{-\pi r^2/t}dt \leq \int_{r/\sqrt{2}}^\infty e^{-2\pi t}dt = \frac{e^{-\pi\sqrt{2}r}}{2\pi}.$$

Both bounds decay faster than any polynomial. Again, we can do a similar analysis for the derivatives and it follows that a is a Schwartz function. Thus, we can take the Fourier transform of a inside of the integral and the identity $a=\hat{a}$ follows from what was observed in the previous section.

From 6.4 we can also see that the for $r>\sqrt{2}$, a can be written as

$$a(r) = \sin^2(\pi r^2/2)\int_0^\infty g_+(it)e^{-\pi tr^2}dt. \tag{6.8}$$

Indeed, the only thing left to justify from the previous section is that one can make the contour shift. We already observed that g_+ is holomorphic and, from expansion 6.4 if follows that

$$|g_+(it)e^{-\pi tr^2}| \to 0,$$

for $r^2>2$, as $t\to\infty$, justifying the contour shift.

Finally, we write a in such a way that we can compute its value at 0 and at $\sqrt{2}$:

Proposition 6.2. *For all $r\in\mathbb{R}$, it holds that*

$$\begin{aligned}a(r) = \sin^2(\pi r^2/2)\Bigg(&\frac{1}{4\pi^2r^4} - \frac{42}{5\pi^2r^2} - \frac{1}{60\pi^2(r^2-2)} \\ &+ \int_0^\infty \left(g_+(it) - 4t + \frac{42}{5\pi} + \frac{e^{2\pi t}}{60\pi}\right)e^{-\pi r^2t}dt\Bigg).\end{aligned} \tag{6.9}$$

Proof. From 6.4 we have that $g_+(it) = 4t + \frac{42}{5\pi} + \frac{e^{2\pi t}}{60\pi} + O(t^2 e^{-2\pi t})$, as $t \to \infty$, so the integral on the right hand side of equality 6.9 converges absolutely for r. For $r^2 > 2$, it also holds that

$$\int_0^\infty \left(4t - \frac{42}{5\pi} - \frac{e^{2\pi t}}{60\pi}\right) e^{-\pi r^2 t} dt = \frac{1}{4\pi^2 r^4} - \frac{42}{5\pi^2 r^2} - \frac{1}{60\pi^2(r^2-2)}. \tag{6.10}$$

Thus, from 6.8 we know that the identity holds for $r^2 > 2$. Furthermore, we can see from that a extends to a holomorphic function in a neighbourhood of $[0, \infty)$ and so does the function on the right hand side of 6.9. By analytic continuation, the identity must hold for every $r \in \mathbb{R}$. □

Just by expanding in Taylor series the sine factor, we conclude that $a(0) = a(\sqrt{2}) = 0$ and $a'(\sqrt{2}) = -\frac{1}{60\sqrt{2}}$.

Doing a similar analysis for the function b, we conclude that it also is a Schwartz function. b can be written as the usual sine factor times the Laplace transform of the function g_- defined as follows

$$g_- = -\frac{1}{120\pi} \frac{5\theta_{00}^8 \theta_{00}^{12} - 5\theta_{00}^4 \theta_{01}^{16} + \theta_{01}^{20}}{\Delta}.$$

g_- can be expanded in Fourier series as follows

$$g_-(z) = -\frac{1}{60\pi} q^{-1} - \frac{12}{5\pi} + \frac{256}{3\pi} q^{1/2} - \frac{5877}{5\pi} q^{1/2} + \dots \tag{6.11}$$

and, if we write $\phi_-(z) = z^2 g_-(-1/z)$, then

$$\phi_-(z) = \frac{512}{3\pi} q^{1/2} + \frac{104448}{5\pi} q^{3/2} + \frac{4027392}{5\pi} q^{5/2} + \dots. \tag{6.12}$$

The Fourier coefficients of g_-, $c_{g_-}(n)$, are also bounded by

$$|c_{g_-}(n)| \leq \frac{1}{30\pi} e^{4\pi\sqrt{n}}, \tag{6.13}$$

for $n \geq 6$. For $r^2 > 2$ it holds that

$$b(r) = \sin^2(\pi r^2/2) \int_0^\infty g_-(it) e^{-\pi r^2 t} dt,$$

and for all $r \in \mathbb{R}$, we can write

$$b(r) = \sin^2(\pi r^2/2) \left(-\frac{12}{5\pi^2 r^2} - \frac{1}{60\pi^2(r^2-2)} + \int_0^\infty \left(g_-(it) + \frac{12}{5\pi} + \frac{e^{2\pi t}}{60\pi}\right) e^{-\pi r^2 t} dt\right). \tag{6.14}$$

Expanding the sine in Taylor series in 6.14 we obtain that $b(0) = b(\sqrt{2}) = 0$ and $b'(\sqrt{2}) = -\frac{1}{60\sqrt{2}}$.

We now set $f = a + b$ and check that is it indeed the magic function that solves the sphere packing problem in dimension 8. One can observe that $f(0) = a(0) + b(0) = 1$ and $\hat{f}(0) = a(0) - b(0) = 1$, as we wanted. The root conditions that the function f would satisfy if it was the magic function also hold: $f(\sqrt{2n}) = a(\sqrt{2n}) + b(\sqrt{2n}) = 0$ for all $n \geq 1$ and $f'(\sqrt{2n}) = a'(\sqrt{2n}) + b'(\sqrt{2n}) = 0$ for all $n \geq 2$; for all $\hat{f}$ we have the same roots stated above and it also holds that $\hat{f}'(\sqrt{2}) = \hat{a}'(\sqrt{2}) - \hat{b}'(\sqrt{2}) = -\frac{1}{60\sqrt{2}} + \frac{1}{60\sqrt{2}} = 0$. The remaining requirements to examine are the sign conditions.

6.2 Computational assistance

Recall that we want to prove that $f(x) \leq 0$ for all $|x|^2 > 2$, and that $\hat{f}(x) \geq 0$ for all $x \in \mathbb{R}^8$. To prove that $f(x) \leq 0$ for all $x \in \mathbb{R}^8$, we let $A(it) = g_+(it) + g_-(it)$. Then, we know that, for $|x|^2 > 2$

$$f(x) = \sin^2(\pi|x|^2/2) \int_0^\infty A(it) e^{-\pi|x|^2 t} dt. \tag{6.15}$$

Observe that if we show that $A(it) < 0$ for all $t \in (0, \infty)$, then the desired inequality follows.

To show that, we first truncate the Fourier expansions 6.2, 6.4, 6.11 and 6.12 and define $A_\infty^{(n)}(it)$ and $A_0^{(n)}(it)$ as the expressions such that

$$\begin{aligned} A(it) &= A_\infty^{(n)}(it) + O(t^2 e^{-\pi n t}), \text{ as } t \to \infty \\ A(it) &= A_0^{(n)}(it) + O(t^2 e^{-\pi n/t}), \text{ as } t \to 0. \end{aligned} \tag{6.16}$$

From 6.1 and 6.13 we get that, for $n \geq 6$

$$\begin{aligned} |A(it) - A_\infty^{(n)}(it)| &\leq \frac{1}{30\pi}(t^2 + t + 1) \sum_{n=m}^{\infty} e^{2\sqrt{2}\pi\sqrt{n}} e^{-\pi n t} \\ |A(it) - A_0^{(n)}(it)| &\leq \frac{1}{30\pi}(t^2 + 1) \sum_{n=m}^{\infty} e^{2\sqrt{2}\pi\sqrt{n}} e^{-\pi n/t}. \end{aligned}$$

We call the expressions on the right hand side $R_\infty^{(n)}$ and $R_0^{(n)}$, respectively. It turns out that one can show computationally that

$$\begin{aligned} &|R_0^{(6)}| \leq |A_0^{(6)}| \text{ for } t \in (0, 1] \\ &|R_\infty^{(6)}| \leq |A_\infty^{(6)}| \text{ for } t \in [1, \infty) \\ &A_0^{(6)}(it) < 0 \text{ for } t \in (0, 1] \\ &A_\infty^{(6)}(it) < 0 \text{ for } t \in [1, \infty), \end{aligned}$$

which proves the result.

To show that $\hat{f}(x) \leq 0$ we do something analogous. Recall that we showed that

$$\hat{f}(x) = \sin^2(\pi|x|^2/2) \int_0^\infty (g_+(it) - g_-(it)) e^{-\pi|x|^2 t} dt \tag{6.17}$$

holds for $|x|^2 > 2$. The expression actually holds for all $x \in \mathbb{R}^8$, since the q^{-1} terms in the Fourier expression of g_+ and g_- cancel out. We define $B = g_+ - g_-$ and quantities $B_\infty^{(n)}$ and $B_0^{(n)}$ as we did in 6.16. The result $B(it) > 0$ follows as before, using computational assistance.

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